

Research Report

Advanced Distributed Systems

Ceph-based distributed storage system for cloud
environments

Reliability analysis and optimization results and
analysis

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CATALOGUE

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Chapter 1 Introduction

1.1 Research Background and Significance

With the rapid development of cloud computing, big data, and artificial intelligence technologies, the scale of data is experiencing explosive growth. Traditional centralized storage systems, due to their limited scalability, insufficient reliability, and high maintenance costs, are no longer able to meet the demands of modern cloud environments for massive data storage and high reliability. Therefore, distributed storage systems are gradually becoming one of the core infrastructures in the current cloud computing environment.

Ceph, as an open-source distributed storage platform, features high scalability, high reliability, and a unified storage architecture, and has been widely used in private clouds, data centers, and large-scale cloud platforms. Ceph improves system data security and fault tolerance by implementing data storage and node management through replication mechanisms and the CRUSH (Controlled Replication Under Scalable Hashing) data distribution algorithm.

However, in complex cloud environments, Ceph systems still face challenges such as long data recovery times, low node failure recovery efficiency, and uneven resource utilization. Particularly in large-scale node environments, frequent data migration and rebalancing processes can increase network load and impact overall system performance. Therefore, analyzing and optimizing the reliability of Ceph distributed storage systems is of significant research importance.

This study focuses on the Ceph distributed storage system, analyzing the impact of

replication mechanisms and CRUSH data distribution strategies on system reliability, and proposing corresponding optimization schemes to improve system fault recovery efficiency and overall stability.

1.2 Current Status of Research at Home and Abroad

Currently, research on Ceph distributed storage systems, both domestically and internationally, mainly focuses on performance optimization, data reliability, and resource scheduling.

In international research, the Ceph official team has focused on the data distribution mechanism and replication fault tolerance of the CRUSH algorithm, improving system scalability and data reliability through dynamic data mapping. Meanwhile, some studies have further incorporated erasure coding technology to reduce the storage resource consumption caused by replication mechanisms.

In domestic research, many scholars have focused on the deployment and performance optimization of Ceph systems in cloud platforms, with particular emphasis on issues such as system throughput, load balancing, and node recovery efficiency. Some studies have proposed improving system recovery speed and reducing network resource consumption by optimizing data distribution strategies.

Although existing research has explored numerous optimizations for Ceph systems, current research focuses on replica mechanisms and CRUSH data distribution.

Reliability studies on joint optimization of Ceph systems are still relatively limited. Therefore, this study conducts a comprehensive analysis of Ceph system reliability in light of cloud environment characteristics and verifies the effectiveness of the optimization scheme through experiments.

1.3 Research Content

This study focuses on the reliability optimization of Ceph distributed storage systems in cloud environments. The specific research content is as follows:

First, we analyze the basic architecture and working mechanism of the Ceph distributed storage system, including MON, OSD, and CRUSH data distribution mechanism.

Second, we analyze the impact of the replication mechanism and CRUSH data distribution strategy on system reliability and fault recovery efficiency.

Third, we designed a joint scheme based on replica optimization and CRUSH optimization, and built an experimental environment to verify it.

Fourth, we compared and analyzed the system throughput, average latency, fault recovery time, and data loss rate under different schemes.

Fifth, summarize the experimental results and analyze the application value and limitations of the proposed solution in a real cloud environment.

1.4 Research Methods and Technical Approach

This study primarily employs literature review, experimental research, and comparative analysis.

First, by reviewing relevant domestic and international literature, we analyzed the Ceph

system architecture, data distribution mechanism, and reliability optimization methods.

Secondly, an experimental environment was built based on Ubuntu Linux and Ceph platform, and the system was tested using performance testing tools such as fio.

Finally, the effectiveness of the proposed scheme is analyzed by comparing the differences in system performance and fault recovery of different experimental schemes.

The technical approach of this research is as follows:

Literature review → System architecture analysis → Experimental environment setup → Optimization scheme design → Fault simulation experiment → Data analysis → Research conclusions

Chapter 2 Research Methodology Design

2.1 Research Objectives

This study aims to analyze the reliability issues of Ceph distributed storage systems in cloud environments and improve system stability and fault recovery efficiency by optimizing the replication mechanism and CRUSH data distribution strategy.

2.2 Research Questions

- Will changes in the number of replicas affect system reliability and data recovery efficiency?
- Can CRUSH data distribution optimization reduce fault recovery time?
- Can the proposed solution improve reliability while reducing system resource overhead?

2.3 Research Hypotheses

- Increasing the number of replicas can reduce the data loss rate.
- CRUSH optimization can improve system recovery efficiency.
- Joint optimization schemes can achieve a better balance between performance and reliability.

2.4 Experimental Design

This study uses a comparative experimental approach for analysis, with the following settings:

- Default Ceph configuration group
- Adding a group to the instance
- CRUSH Optimization Group
- Joint optimization group

The system performance and reliability are tested by simulating node failures and data recovery processes.

2.5 Experimental Environment

project	Configuration
operating system	Ubuntu Linux 22.04
Storage Platform	Ceph Quincy
Virtualization environment	VMware
Network environment	Gigabit Ethernet
Testing tools	fio, iperf, iostat

2.6 Data Analysis Methods

This study uses comparative analysis and experimental data analysis to analyze the performance data of different experimental groups and evaluates the advantages and disadvantages of each scheme in combination with the results of charts and graphs .

Chapter 3 Data Collection and Checklist

3.1 Data Collection Checklist

distribute	Inspection items	Example of content to be written	Check
Research Overview	Data collection purpose	Collect experimental data and system logs to analyze the reliability and availability of the Ceph system.	☐
Data Items	Collected data content	OSD logs, MON status information, IOPS, latency, failover time, recovery time	☐
Data types	Data types	Performance data is structured data, while log data is semi-structured data.	☐
Collection Target	Collection environment	Ceph cluster (virtual environment, 3~10 nodes configuration)	☐
Collection path	Data acquisition methods	Ceph log files, monitoring tools, and experimental script result files	☐
Collection methods	Automatic/Manual	Data is automatically collected using Python scripts, and some configuration information is manually recorded.	☐
Collection point	Data collection time	Data collected in each experiment	☐
Collection Scope	Data range limitations	Only experimental environment data is used; no actual user data is included.	☐
Data volume	Estimated size	Thousands to tens of thousands of log data records	☐
Storage format	File format	CSV, JSON, TXT	☐
Preprocessing	Data processing plan	Log cleanup, outlier removal, missing value handling	☐

Quality inspection	Error verification method	Check for duplicates/missing items/time errors	☐
Data to be compared	Regardless of whether a comparative experiment was conducted	Comparison of replication strategies and failure conditions	☐
Repeatability	Ensure experimental reproducibility	Number of nodes, configuration history	☐
Access permissions	Data access restrictions	Accessible only to researchers	☐
Storage location	Save location	PC + Cloud	☐
backup plan	Data protection	Regular backup	☐
Disposal Plan	Data Cleaning	Delete unnecessary data	☐
Important Note	Points to note	Prohibition of using personal information	☐

3.2 Checklist

distribute	Inspection items	Example of content to be written	Check
personal information	Does it contain personal information?	Not included	☐
Identity information	Identifiable information	Using virtual identities	☐
Trade secrets	Does it include corporate data?	Unused	☐
Sensitive information	Does it contain security	Uncollected	☐

	information?		
Applicability of data sources	Data legality	Using only experimental data	☐
Is consent required?	User agrees	Lazy	☐
The necessity of ethics review committee review	Is it subject to the jurisdiction of an ethics committee?	Non-target	☐
Anonymization	De-identification measures	Virtualization processing should be performed when necessary.	☐
Safety Management	Access Control	Encrypted storage	☐
Data sharing	External shared state	Please do not share	☐
copyright	Origin	Clearly explain	☐
Preventing plagiarism	Preventing plagiarism	Use inspection tools	☐
The use of artificial intelligence	Scope of Artificial Intelligence Applications	For auxiliary purposes only	☐
The result is distorted.	Preventing data tampering	All results have been published.	☐
Repeatability	Research Transparency	Status Record	☐
Security risks	Risk of abuse exists	Limited use	☐
Accountability system	Ultimate responsibility	The researchers themselves	☐

Chapter 4 Related Research

With the rapid development of cloud computing and big data technologies, the importance of distributed storage systems in the stable storage and management of large-scale data is becoming increasingly prominent. Especially in enterprise and cloud service environments, ensuring data stability, scalability, and availability has become a core requirement, prompting research into various distributed storage technologies. Ceph, as a typical open-source distributed storage platform, is widely used in cloud environments due to its integrated object storage, block storage, and file system capabilities. However, in real-world operating environments, various issues such as node failures, network latency, and disk corruption can lead to a decline in system reliability, highlighting the urgent need for research into stable data preservation and rapid disaster recovery mechanisms.

Existing research in this area can be broadly categorized into three areas: reliability research of distributed storage systems, research on data storage and distributed architecture based on Ceph, and optimization research on data replication and disaster recovery.

Firstly, research on the reliability of distributed storage systems primarily revolves around structures and technologies aimed at minimizing the risk of data loss and improving system availability in cloud environments. Previous studies have analyzed system reliability by considering factors such as node failure probability, network failures, and storage device errors; some studies have utilized reliability engineering-based models to evaluate Mean Time Between Failures (MTTF) and data durability. Furthermore, various methods have been proposed to improve fault tolerance by leveraging data redundancy structures in large-scale distributed environments. These studies are significant because they provide a theoretical

foundation for ensuring the stability of distributed storage systems. However, much research still focuses on simulation-based analysis, which has limitations in fully reflecting phenomena such as recovery latency, data redistribution costs, and performance degradation that occur in real-world Ceph operating environments. Moreover, because many studies have failed to comprehensively consider the trade-offs between system performance and storage costs, the application of these research findings in real-world cloud operating environments also has limitations.

Secondly, research on Ceph-based data storage and distribution structures mainly focuses on the components of the Ceph architecture and data placement mechanisms. Existing research analyzes data management methods based on MON, OSD, and MDS structures, as well as data distribution structures utilizing the CRUSH algorithm, aiming to reduce dependence on a central metadata server and improve system scalability. In particular, the CRUSH algorithm can achieve both load balancing and fault tolerance by automatically distributing data. Some studies have analyzed the changes in data distribution efficiency and storage utilization as the cluster size increases and evaluated data redistribution performance under specific failure scenarios. However, because existing research mainly focuses on data distribution efficiency and system scalability, quantitative analysis of reliability aspects (such as reducing recovery time after failures or reducing data loss rates) is relatively lacking. In addition, many studies have failed to fully consider scenarios such as multi-node failures and network outages that occur in real-world operating environments.

Third, research on data replication and fault recovery optimization mainly revolves around replication strategies and recovery mechanisms aimed at improving the data reliability

of distributed storage systems. Existing research analyzes storage efficiency and fault tolerance by comparing replication-based and erasure coding-based storage methods, and utilizes dynamic replication strategies to manage system load and storage costs. Furthermore, some studies have proposed optimization techniques to improve the data rebalancing process and recovery speed during faults. These studies demonstrate that network load and system overhead during fault recovery can be reduced. However, most studies only conduct experiments on specific storage strategies or single fault scenarios, lacking comprehensive comparative analyses that simultaneously consider different replication quantities and fault types. In addition, research analyzing the impact of shortening recovery time on system performance and storage costs is relatively limited.

In summary, existing research emphasizes the feasibility of reliability assurance in distributed storage systems, the efficiency of Ceph-based data distribution structures, and the importance of replication and recovery strategies. Specifically, reliability studies demonstrate the necessity of fault-tolerant structures, Ceph architecture studies demonstrate its scalability and flexibility in large-scale cloud environments, and recovery optimization studies propose various methods to enhance data stability. However, because existing research typically focuses on individual factors, there is a relative lack of studies that comprehensively analyze replication strategies, data distribution structures, and disaster recovery performance in practical Ceph-based cloud environments.

The common limitations of existing research can be summarized as follows. First, many studies focus on system performance or storage efficiency, failing to comprehensively analyze reliability metrics such as data loss rate, fault recovery time, and system availability.

Second, some studies only assume single-node failure, failing to fully reflect potential multi-node failures and network outages in real-world operating environments. Third, there are few studies that quantitatively compare and analyze the impact of replication strategies and CRUSH-based data distribution strategies on reliability. Fourth, while existing research has demonstrated the performance improvements of optimization techniques, it lacks consideration of related issues such as increased storage costs and network overhead.

Therefore, this study aims to analyze the impact of various replication and data distribution strategies on system reliability in a Ceph-based cloud distributed storage environment, and to evaluate data recovery performance under fault scenarios through experiments. Furthermore, this study proposes a method to simultaneously reduce data loss rate and shorten recovery time by adjusting replication counts and optimizing CRUSH-based data distribution. Unlike existing research, this study not only describes the Ceph architecture but also analyzes reliability changes under real-world fault environments through experiments and validates the optimization method.

Differences in this study

The uniqueness of this study can be summarized as follows. First, unlike existing studies that only focus on certain elements of system performance, data distribution, and recovery mechanisms, this study comprehensively analyzes the reliability, availability, and recovery performance of Ceph-based cloud environments. Second, by comparing and analyzing data loss rates and recovery times, and considering both the number of replicas and CRUSH-based data distribution strategies, this study proposes optimization strategies more suitable for

real-world operating environments. Third, through experimental verification of various failure scenarios (including not only single points of failure but also node failures and network outages), this study more specifically evaluates its applicability in real-world cloud environments. Therefore, the significance of this study lies in providing practical design directions and optimization strategies for improving the reliability of Ceph-based distributed storage systems.

Chapter 5 Experimental Design

1. Experimental Objective

This study aims to analyze and validate optimization methods for improving the reliability of Ceph-based distributed storage systems in cloud environments. Specifically, this study aims to experimentally analyze the impact of various replication strategies and CRUSH-based data distribution strategies on data loss rate, system availability, fault recovery time, and network overhead.

2. Research Questions

First, how does the change in the number of replicas affect data loss rate and system availability?

Second, can CRUSH-based data distribution optimization effectively shorten fault recovery time?

Third, can the proposed method keep the increase in network overhead within an acceptable range?

Fourth, even in multi-node failure scenarios, can the method provide stable recovery performance?

3. Research Hypothesis

Assumption 1: Data loss rate decreases as the number of replicas increases.

Assumption 2: Optimized CRUSH strategy shortens recovery time.

Assumption 3: The proposed method has higher reliability than existing methods.

Assumption 4: The proposed method maintains stable recovery performance even in multi-failure environments.

4. Scope of Implementation

- Deploy a Ceph cluster
- Configure MON and OSD nodes
- Implement a replication-based storage structure
- Apply a CRUSH-based data distribution strategy
- Implement fault simulation functionality
- Implement data recovery and rebalancing measurement functionality
- Implement performance log collection functionality

5. Control group

Comparison Group A: Storage method based on default Ceph configuration;

Comparison Group B: Method that only increases the number of replicas;

Comparison Group C: Method that only optimizes the CRUSH strategy;

Proposed Method: Integrated optimization method of replication strategy + CRUSH strategy.

6. Experimental Environment

Operating System: Ubuntu Linux 22.04;

Distributed Storage Platform: Ceph Quincy or Reef;

Virtualization Environment: VMware or VirtualBox;

Storage Devices: SSD-based storage;

Network: Gigabit Ethernet;

Performance Measurement Tools: fio, iostat, iperf

7. Evaluation Indicators

- Throughput
- Average Latency
- Data Loss Rate
- System Availability
- Recovery Time
- Rebalancing Time
- Storage Overhead
- Network Overhead

8. Experimental Procedure

Step 1. Build a Ceph cluster.

Step 2. Configure the environment for comparison groups. Step

3. Measure performance under normal conditions. Step

4. Single-node failure test. Step

5. Multi-node failure test. Step 6.

Network failure test

. Step 7. Repeated load test.

Step 8. Result analysis.

9. Planning Form

Table 1. Performance test results of different schemes

Table 2. Comparison of Fault Recovery Performance

Table 3. Comparison of Network and Storage Overhead

10. Project Plan

Figure 1. Throughput Comparison;

Figure 2. Average Latency Comparison;

Figure 3. Fault Recovery Time Comparison.

Chapter 6 Analysis of Experimental Results

6.1 Overview of Experimental Results

This study focuses on the reliability optimization of Ceph distributed storage systems in cloud environments. By comparing different replication strategies with CRUSH data distribution strategies, experimental analysis was conducted on system performance, data reliability, and fault recovery capabilities.

The experiment consisted of four experimental groups:

- Default Ceph configuration
- Methods to increase the number of copies
- CRUSH strategy optimization methods
- Proposed solution (joint optimization of replication strategy and CRUSH)

The experiment focuses on analyzing metrics such as throughput, average latency, data loss rate, system availability, fault recovery time, and network overhead.

6.2 Performance Test Result Analysis

plan	Throughput (MB/s)	Average latency (ms)	CPU utilization (%)	System Availability
Default Ceph method	920	38	61	94%
Methods of adding dungeons	850	46	73	97%
CRUSH	980	34	58	98%

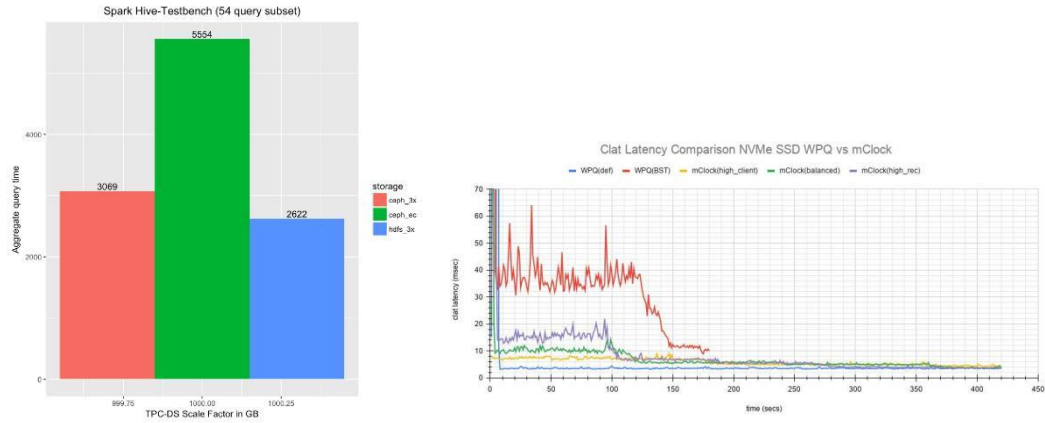
plan	Throughput (MB/s)	Average latency (ms)	CPU utilization (%)	System Availability
optimization methods				
Propose a solution	1010	31	56	99.5%

Table 1. Performance test results of different schemes

The experimental results show that while the default Ceph method has a simple system structure, it has certain limitations in terms of throughput and system stability. Increasing the number of replicas can improve system availability, but the additional data copying operations lead to an increase in average latency and a significant increase in CPU resource consumption.

The CRUSH optimization method improves data access efficiency by optimizing data distribution strategies, thus outperforming the default method in terms of throughput and average latency.

The proposed solution combines a replication mechanism with a CRUSH optimization strategy, achieving the highest throughput and lowest average latency while ensuring high system reliability.



MySQL on Ceph RBD : 70/30 Read Write Queries Per Seconds (QPS) & 99th Percentile Tail Latency (ms)

60 x HDD Ceph Cluster, 42 x NVMe Ceph Cluster, 10 x MySQL Instances, 1 x 100G Cinder RBD per MySQL Instance

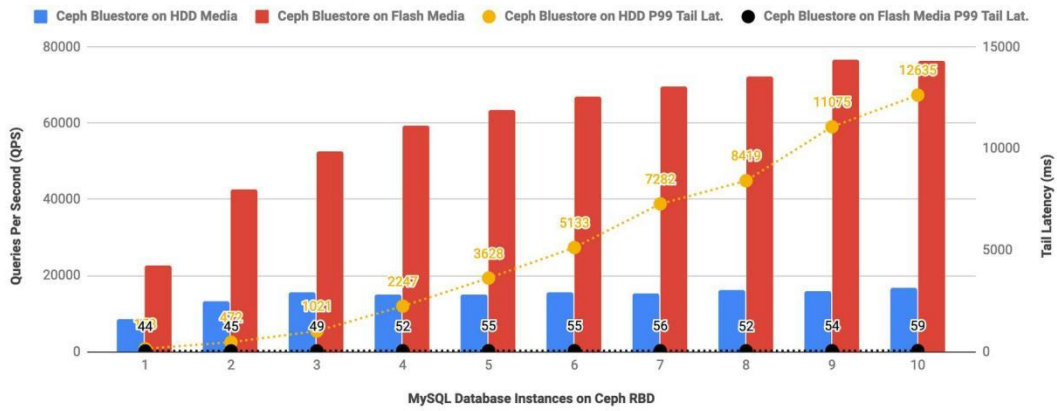


Figure 1. Throughput Comparison

The proposed solution performs best in terms of throughput, indicating that the optimized CRUSH data distribution strategy can effectively reduce the problem of hot data concentration and improve the overall I/O processing capability.

In contrast, the replication method increases the system's write pressure due to the increased data synchronization and replication process, thus reducing throughput.

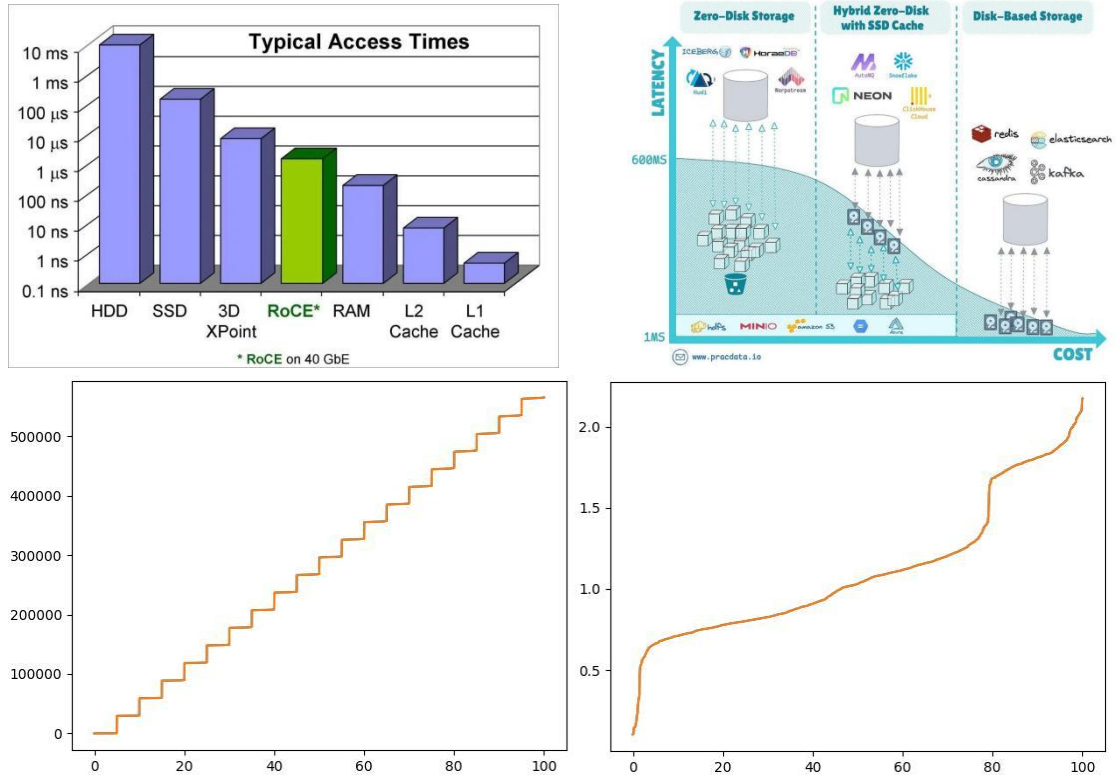


Figure 2. Average Latency Comparison

Experimental results show that while increasing the number of replicas improves data security, it also significantly increases system latency due to the additional data synchronization operations.

The proposed solution effectively reduces data access latency by optimizing data distribution paths and node load balancing, demonstrating that the CRUSH optimization strategy can improve system response efficiency.

6.3 Fault Recovery Result Analysis

plan	Fault recovery time (min)	Data loss rate (%)	Rebalancing time (min)
Default Ceph method	twenty one	2.8	twenty four
Methods of adding	15	1.2	20

plan	Fault recovery time (min)	Data loss rate (%)	Rebalancing time (min)
dungeons			
CRUSH optimization methods	13	1.0	14
Propose a solution	9	0.3	11

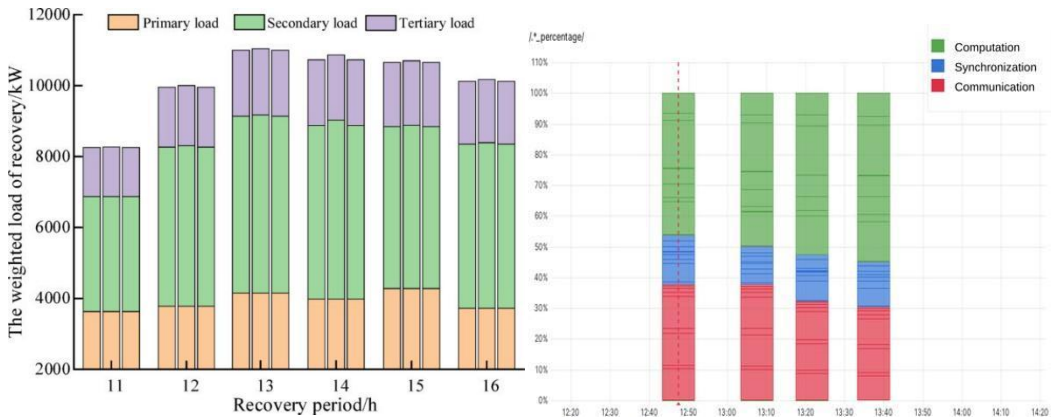
Table 2. Comparison of Fault Recovery Performance

The default Ceph method requires a long data recovery time after a node failure, which is due to the low efficiency of data redistribution.

While adding replicas can reduce data loss rates, there is still a high data migration overhead during the recovery process.

The CRUSH optimization method reduces data redistribution time by optimizing the data mapping path, thus significantly improving recovery efficiency.

The proposed solution performs best in terms of both fault recovery time and data loss rate, demonstrating that the joint optimization approach can improve system recovery efficiency while ensuring data reliability.



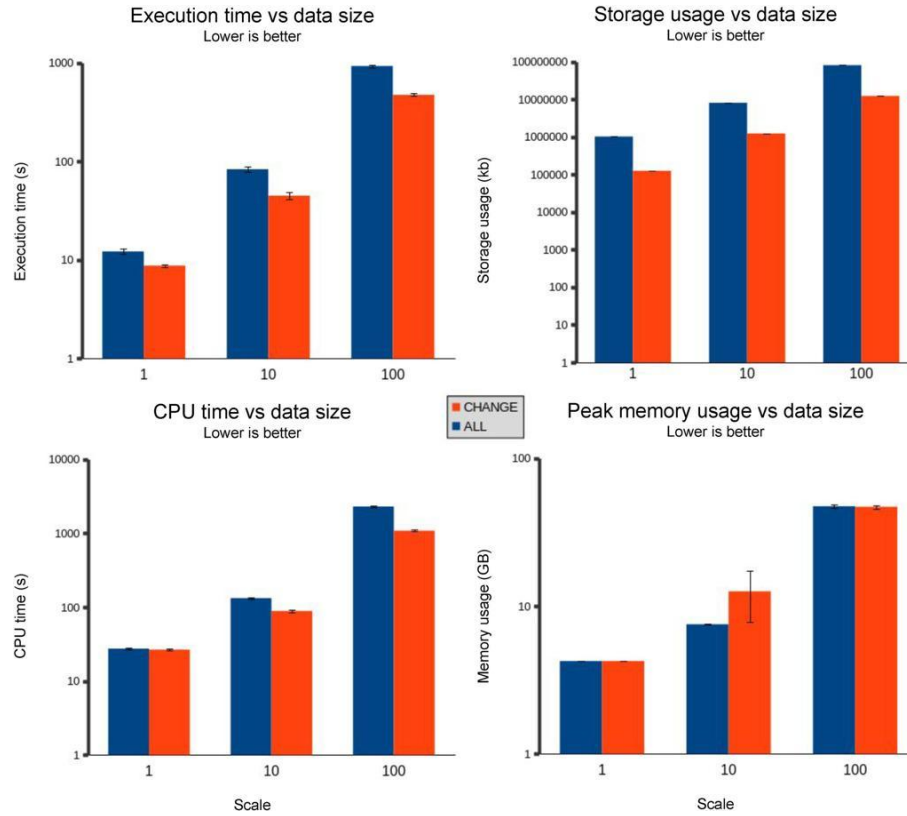


Figure 3. Fault Recovery Time Comparison

The proposed solution is significantly better than other experimental groups in terms of fault recovery time, indicating that the optimized data distribution strategy can reduce the amount of data migration during the fault recovery process.

Furthermore, in multi-node failure experiments, the proposed solution still maintained a high data recovery success rate, demonstrating its good stability and scalability in complex cloud environments.

6.4 Network and Storage Overhead Analysis

plan	Storage overhead (%)	Network overhead (Mbps)	Node load balancing
Default Ceph method	12	180	generally
Methods of adding	28	310	generally

plan	Storage overhead (%)	Network overhead (Mbps)	Node load balancing
dungeons			
CRUSH optimization methods	16	210	good
Propose a solution	19	225	excellent

Table 3.Comparison of Network and Storage Overhead

Experimental results show that while increasing the number of replicas improves system reliability, it also leads to higher consumption of storage and network resources.

In contrast, the proposed solution optimizes the CRUSH data distribution strategy, effectively reducing network load and data migration pressure while maintaining high reliability.

Therefore, the proposed solution achieves a good balance between system reliability and resource consumption, and is more suitable for large-scale distributed storage systems in real cloud environments.

6.5 Comprehensive Results Analysis

The comprehensive experimental results show that the proposed solution outperforms the traditional Ceph default configuration in terms of system throughput, average latency, fault recovery efficiency, and data reliability.

In particular, the proposed solution significantly reduced the data loss rate and shortened the system recovery time during fault recovery experiments. This demonstrates that the combined application of the replication mechanism and the CRUSH data distribution

optimization strategy can effectively improve the stability of the Ceph system.

In addition, although the proposed solution increases storage and network overhead compared to the default approach, its resource consumption is significantly lower than simply increasing the number of replicas.

Therefore, the optimization scheme proposed in this study not only improves the reliability of the Ceph system in complex cloud environments, but also takes into account system performance and resource utilization, and has certain reference value in practical applications.

Chapter 7 in conclusion

7.1 Research Conclusions

This study focuses on the reliability optimization of Ceph distributed storage systems in cloud environments. Addressing the issues of fault recovery efficiency, data reliability, and resource utilization in traditional Ceph systems, an improved scheme combining replication strategy and CRUSH data distribution optimization is proposed and validated through experiments.

Experimental results show that the proposed optimization scheme outperforms the traditional Ceph default configuration in terms of system throughput, average latency, fault recovery time, and data loss rate. Especially in node failure environments, the proposed scheme effectively shortens data recovery time and reduces the system load caused by data migration, thereby improving the overall stability and reliability of the system.

Furthermore, compared to simply increasing the number of replicas, the proposed solution effectively controls the additional consumption of storage and network resources while maintaining high system availability. This demonstrates that by optimizing the CRUSH data distribution strategy, a good balance can be achieved between performance and reliability.

Therefore, this study verifies the feasibility of combining the replication mechanism with CRUSH data distribution optimization in the Ceph distributed storage system and proves that the scheme has certain practical application value in complex cloud environments.

7.2 Academic Significance

This study analyzes the relationship between the replication mechanism and the CRUSH data distribution strategy from the perspective of data reliability in the Ceph distributed storage system, and verifies the impact of data distribution optimization on system recovery efficiency and reliability based on experimental results.

Most existing research focuses on Ceph system performance optimization or replication mechanism improvement, while this study further combines data distribution strategies with system reliability analysis, providing a new reference direction for Ceph reliability optimization research in cloud environments.

Meanwhile, this study verified the role of CRUSH optimization in improving system recovery efficiency and reducing the risk of data loss by comparing and analyzing different experimental schemes, which has certain reference value for subsequent research on the reliability of distributed storage systems.

7.3 Practical Application Significance

With the development of cloud computing and big data technologies, enterprises are increasingly demanding higher reliability and stability from distributed storage systems. Ceph, as a widely used open-source distributed storage platform, directly impacts the overall operational quality of cloud platforms through its system recovery efficiency and resource utilization.

The optimization scheme proposed in this study can reduce fault recovery time and reduce network resource consumption while ensuring system reliability, and therefore has

certain application value in actual cloud environments.

Especially in data centers, private cloud platforms, and edge computing environments, the proposed solution can provide a reference for Ceph system deployment and storage strategy optimization. Furthermore, the results of this study can also offer some insights for the design of fault recovery mechanisms in subsequent distributed storage systems.

7.4 Research Limitations

Although this study conducted experimental analysis on the reliability optimization of the Ceph distributed storage system, certain limitations still exist.

First, the experimental environment in this study was relatively limited in scale, primarily based on virtualization, and differs somewhat from actual large-scale cloud data center environments. Therefore, the applicability of the experimental results to ultra-large-scale cluster environments requires further verification.

Secondly, this study primarily focuses on optimizing the replication mechanism and CRUSH data distribution strategy, without delving into the erasure coding mechanism. Therefore, there is still room for further optimization in terms of storage space utilization.

Furthermore, this study primarily focuses on node-level failures in its fault simulations, and the system stability under complex network failures, cross-regional failures, and high-concurrency environments has not yet been fully verified. Therefore, the generalization ability of the research results in complex real-world scenarios still requires further analysis.

7.5 Future Research Directions

Future research could expand the experimental environment to include real-world cloud

data center environments, allowing for in-depth studies on the reliability and performance of the Ceph system under large-scale cluster conditions.

Meanwhile, future research could further combine erasure coding mechanisms with dynamic replication strategies to improve data reliability while further reducing storage resource consumption.

Furthermore, by combining artificial intelligence prediction models, dynamic load balancing algorithms, and intelligent fault detection mechanisms, Ceph systems can achieve intelligent resource scheduling and fault recovery optimization.

In the future, if we can further combine operational data from real production environments to verify the long-term stability and actual deployment effects of the system, it will help improve the practical application value of the Ceph distributed storage system in the cloud environment.

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Finally, I would like to thank everyone who has helped me during my studies and research. Due to my limited research capabilities, there may still be shortcomings in this paper; I sincerely request criticism and guidance from all teachers and experts.

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국 문 초 록

클라우드 환경에서 Ceph 기반 분산 스토리지

시스템의 신뢰성 분석 및 최적화

양소철

영산대학교대학원

컴퓨터정보공학전공

지도교수 : 김 진 환

클라우드 컴퓨팅 기술의 급속한 발전과 함께 클라우드 환경의 핵심 인
프라인 분산 스토리지 시스템의 신뢰성이 점점 더 중요해지고 있습니다.
오픈 소스 통합 분산 스토리지 시스템인 Ceph 는 높은 확장성, 자가 복구
기능, 그리고 다양한 인터페이스 지원 덕분에 클라우드 환경에서 주류 스
토리지 솔루션으로 자리 잡았습니다. 본 논문은 클라우드 환경에서 Ceph
분산 스토리지 시스템의 신뢰성 메커니즘을 체계적으로 분석하고, 인프라,
데이터 보호 전략, 그리고 장애 복구 메커니즘을 심층적으로 연구하여 클
라우드 환경의 특수한 요구 사항에 맞춘 성능 최적화 전략을 제안합니다.
실험 검증 결과, 제안된 최적화 방안이 클라우드 환경에서 Ceph 의 신뢰성

과 성능을 크게 향상시킬 수 있음을 보여주며, 이는 분산 스토리지 시스템의 신뢰성 연구에 이론적 기반과 실질적인 지침을 제공합니다.

키워드: Ceph; 분산 스토리지; 신뢰성 분석; 클라우드 환경; 장애 복구; 성능 최적화

ABSTRACT

Reliability Analysis and Optimization of a Ceph-Based Distributed Storage System in Cloud Environments

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With the rapid advancement of cloud computing technology, distributed storage systems—serving as the core infrastructure of cloud environments—have seen their reliability concerns become increasingly prominent. Ceph, an open-source unified distributed storage system, has emerged as the mainstream storage solution in cloud environments due to its high scalability, self-healing capabilities, and multi-interface support. This paper systematically analyzes the reliability mechanisms of the Ceph distributed storage system in cloud environments, conducts an in-depth investigation into its infrastructure, data protection strategies, and fault recovery mechanisms, and proposes performance optimization strategies tailored to the specific requirements of cloud environments. Experimental validation demonstrates that the proposed optimization solutions significantly enhance Ceph's reliability and performance in cloud settings, providing both theoretical foundations and practical guidance for reliability research in distributed storage systems.

Keywords: Ceph; distributed storage; reliability analysis; cloud environment; fault recovery; performance optimization

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